

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

After reviewing the organization's vulnerabilities, I recommend the following network hardening methods:

1. Multifactor Authentication (MFA)
 - Implement MFA for all employee and administrator accounts.
 - MFA requires users to verify their identity using an additional authentication factor, such as a code sent to a mobile device.
2. Strong Password Policy
 - Require employees to create strong, unique passwords and prohibit password sharing.
 - Change the default database administrator password immediately and enforce password complexity requirements.
3. Firewall Rule Configuration
 - Configure firewall rules to filter incoming and outgoing traffic.
 - Allow only authorized network traffic and block unnecessary ports and connections.

Part 2: Explain your recommendations

Multifactor Authentication is effective because it adds an additional layer of security beyond a password. Even if an attacker obtains a user's password, they would still need the second authentication factor to gain access. MFA should be enabled continuously and reviewed periodically.

A strong password policy is effective because it reduces the likelihood of attackers successfully guessing passwords through brute force attacks. Password sharing should be prohibited, and employees should receive regular security awareness training. Password policies should be enforced at all times

and reviewed regularly.

Firewall rule configuration is effective because it helps control the traffic entering and leaving the network. Properly configured firewalls can block unauthorized access attempts and reduce exposure to network-based attacks. Firewall rules should be monitored continuously and reviewed whenever network changes occur.

Together, these hardening methods address the organization's identified vulnerabilities and significantly reduce the risk of future data breaches.